

Variance Reduction for Monte Carlo Pricing of Credit Derivatives

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Abstract

This project investigates Monte Carlo variance reduction techniques in the context of credit derivatives, using stochastic hazard rates under an HJM framework and a range-accrual credit-linked note (CLN) as the test payoff. The goal is to build a clear and reproducible

numerical pipeline that prices credit products under a dynamic term structure and evaluates how common variance reduction tools perform when applied to structured credit payoffs.

The work begins by constructing deterministic short rate and hazard curves, then embedding these curves inside a discrete HJM hazard model that produces arbitrage-free forward hazard surfaces. A smooth pathwise survival approach is used instead of explicit default simulation, which stabilises Monte Carlo convergence for path-dependent credit products. The exotic CLN payoff is coded in this framework and used to test the effectiveness of antithetic variates and control variates.

Results show that the simulation engine correctly reprices the initial term structures and produces stable Monte Carlo estimates for both investment grade and high yield names. Antithetic variates consistently provide strong variance reduction, while the control variate gives only modest improvement due to weaker realised correlation with the exotic payoff. Overall, the project demonstrates that simple variance reduction tools can materially improve efficiency when pricing structured credit products by simulation.

1 Introduction

This introduction outlines the project goals, constraints, and contribution. The project is a tutorial on Monte Carlo variance reduction with credit as an application domain. It motivates why stochastic hazards and exotic features push us to simulation, and it previews the toolkit and results.

2 Credit Model Setup

Note on Sources. The development in this section follows the presentation in *Credit Derivatives: Pricing Models, Models, Pricing, and Implementation* by [Schönbucher(2003)] very closely. The notation has been adapted to fit the structure and objectives of this project, but the conceptual framework and modelling sequence are based on Schoenbucher's exposition.

2.1 Credit Product Definitions

This section introduces three core instruments used throughout the project: risky bonds, credit default swaps, and credit linked notes. The goal is to give a clear and intuitive explanation that assumes no prior knowledge of credit markets. Credit markets are more complex and nuanced than what we have presented below. However, we opt for simple models that lead to more meaningful understanding, instead of capturing all the economic realities exactly. We first explain how default works, then show how discount factors and survival probabilities enter the valuation of each cashflow. We break each payoff into simple components and give diagrams that illustrate the structure visually.

Default, Recovery, and Survival

A borrower may default at any time before a bond matures. Default means:

- all future coupon payments stop,
- the principal is not paid back in full,
- instead the investor receives a recovery amount R times notional,
- recovery is typically paid immediately when default occurs.

If default happens at time t , this event has probability density $h(t)S(0, t)$, where $h(t)$ is the hazard rate and $S(0, t)$ is the survival probability up to t .

Discount factors $D(0, t)$ reduce future cashflows to present value. Survival probabilities $S(0, t)$ weigh cashflows by the chance that the issuer has not defaulted.

Risky Bonds

A risky bond pays periodic coupons and a principal at maturity, but only if the issuer survives until these dates. If the issuer defaults earlier, coupons stop and the investor receives a recovery payment. We decompose the value of a risky bond into three parts:

1. Coupon leg

$$V_{\text{coupon}} = \sum_i c D(0, t_i) S(0, t_i).$$

2. Redemption leg

$$V_{\text{redemption}} = D(0, T) S(0, T).$$

3. Recovery leg

$$V_{\text{recovery}} = \int_0^T R D(0, t) h(t) S(0, t) dt.$$

Total value

$$V_{\text{risky}} = V_{\text{coupon}} + V_{\text{redemption}} + V_{\text{recovery}}.$$

Note that the coupon and redemption payments occur on fixed dates, while recovery can occur at any time; the time of default is random. Hence, V_{coupon} is a summation and V_{recovery} is an integral. In this tutorial, we ignore issues around accrual on default.

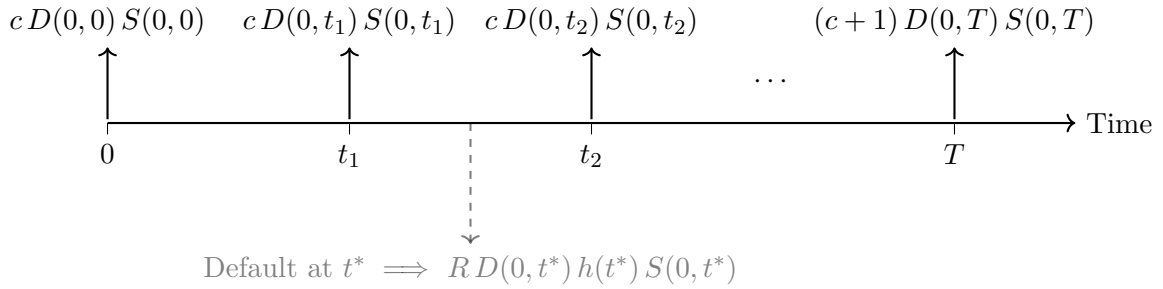


Figure 1: Risky bond cashflows. Coupons c are paid until maturity T if no default occurs. Default at t^* leads to recovery R and termination of future payments. Coupon, Redemption, and Recovery payments are scaled by the discount factors and survival probabilities

Credit Default Swaps

A credit default swap (CDS) transfers the default risk of a reference entity from one party to another. If default occurs before the maturity of the swap, premium payments stop, and the party paying the premium and receives $1 - R$. There are two legs.

Premium leg

$$V_{\text{prem}} = s \sum_i D(0, t_i) S(0, t_i).$$

Protection leg

$$V_{\text{prot}} = (1 - R) \int_0^T D(0, t) h(t) S(0, t) dt.$$

The fair spread s is the spread such that $V_{\text{prem}} = V_{\text{prot}}$ at the start of the swap. We ignore any complexities concerning upfront payments.

Intuition

- A protection buyer (the party *paying* the spread s) is short credit risk.
- A protection seller (the party *receiving* the spread s) is long credit risk.

Identity Linking Bonds and CDS

$$\text{Risk-Free Bond} = \text{Risky Bond} + \text{CDS}.$$

This identity shows how the CDS removes the risky bond's default component.

Credit Linked Notes

A credit linked note (CLN) embeds a short CDS position into a funded bond:

$$\text{CLN} = \text{Risk-Free Bond} - \text{CDS}.$$

The coupon is higher because the investor is selling credit protection. They are receiving the risk-free yield plus the additional yield from being long credit risk. CLNs allow institutions that cannot trade derivatives to take credit risk in a funded format. Unfunded derivatives like a CDS do not require upfront payment, and exchange only regular premium and protection payments. Funded derivatives like credit linked notes require an initial investment because the credit exposure is embedded in a bond structure.

2.2 Deterministic Hazard and Short Rate Curves

In a general reduced form model, the short rate $r(t)$ and hazard rate $h(t)$ are stochastic processes. We define the pathwise discount and survival factors

$$Q_r(0, t) := \exp\left(-\int_0^t r(u) du\right), \quad Q_h(0, t) := \exp\left(-\int_0^t h(u) du\right).$$

The time-zero discount factor and survival probability are then

$$D(0, t) = \mathbb{E}[Q_r(0, t)], \quad S(0, t) = \mathbb{E}[Q_h(0, t)],$$

where the expectation is taken under the risk neutral measure.

If the short rate and hazard rate are deterministic curves, then the expectations are trivial and we can write

$$D(0, t) = \exp\left(-\int_0^t r(u) du\right), \quad S(0, t) = \exp\left(-\int_0^t h(u) du\right).$$

We avoid market calibration and instead specify smooth parametric curves $r(t)$ and $h(t)$ with closed form integrals. This makes $D(0, t)$ and $S(0, t)$ easy to compute, and simulates cleanly within the HJM framework to be described in detail in [section].

2.3 Exponential Hazard and Short Rate Curves

$$r(t) = a_r + b_r e^{-c_r t}, \quad h(t) = a_h + b_h e^{-c_h t}.$$

Integrated forms:

$$Q(0, t) = \exp\left(-a_r t - \frac{b_r}{c_r}(1 - e^{-c_r t})\right),$$
$$S(0, t) = \exp\left(-a_h t - \frac{b_h}{c_h}(1 - e^{-c_h t})\right).$$

These deterministic curves will become the *initial conditions* for the HJM model.

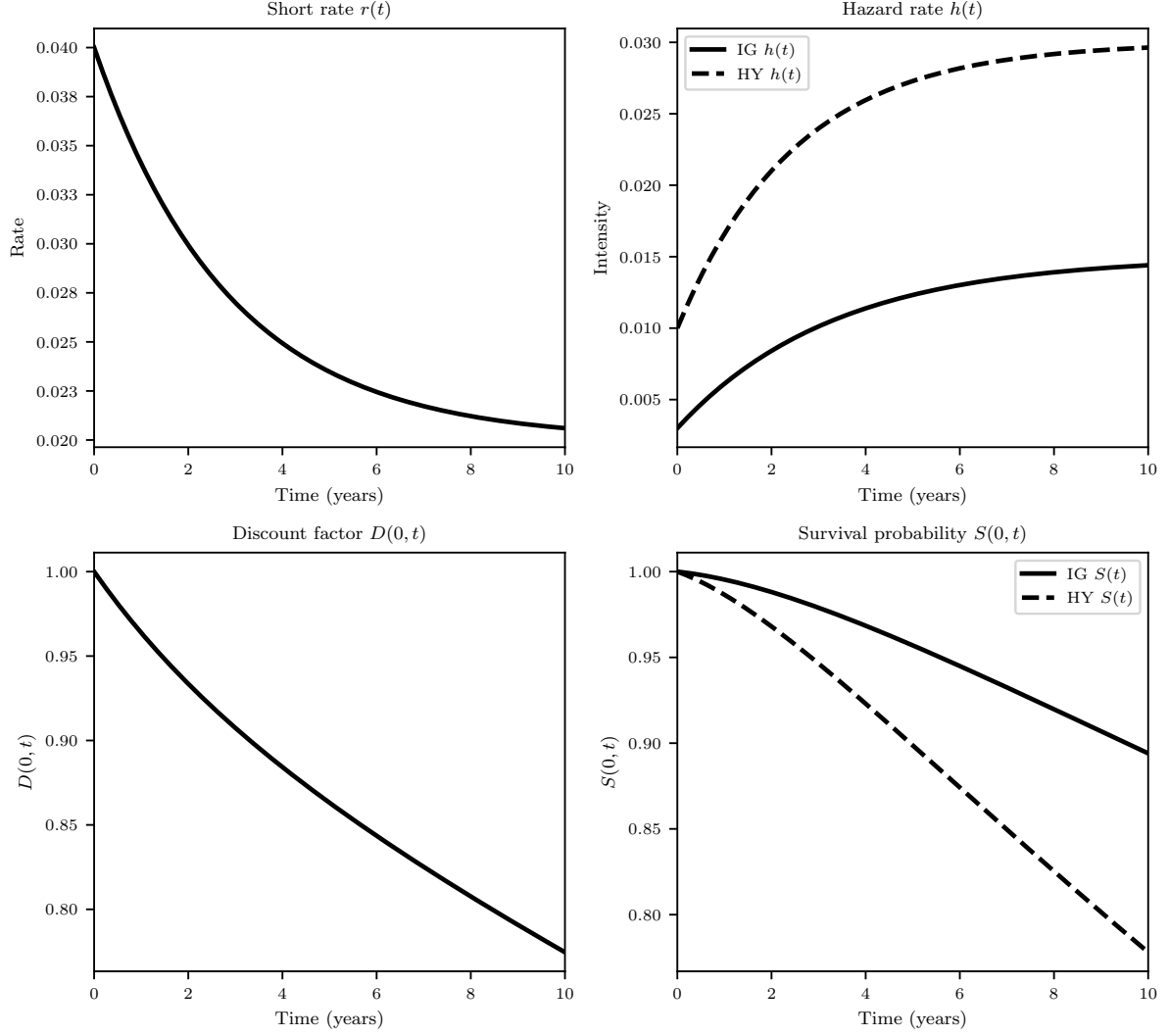


Figure 2: Short rates, hazard rates, discount factors, and survival probabilities.

Interpretation of the Parameters

- Long-run level a : sets the asymptotic intensity/rate.
- Shift b : controls slope at $t = 0$.
- Speed c : determines exponential decay.
 - Equivalently the half-life: $t_{1/2} = \frac{\ln 2}{c}$.

Chosen Parameters and Economic Motivation

The project uses simple but realistic parameter sets for the short rate and for two credit qualities: an investment grade (IG) name and a high yield (HY) name. These curves are chosen for plausibility rather than calibration but retain appropriate risk levels and term structure slopes.

Short rate parameters

$$a_r = 0.02, \quad b_r = 0.02, \quad c_r = 0.35.$$

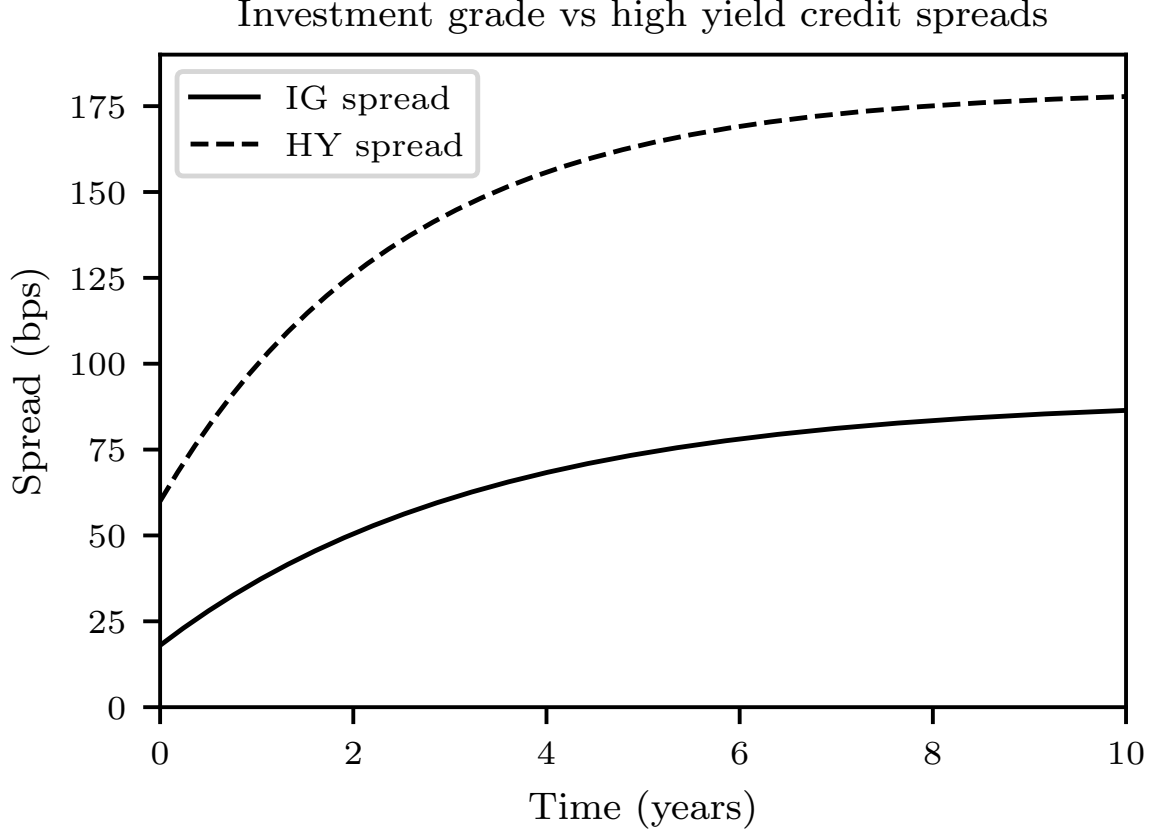


Figure 3: Investment grade and high yield credit spreads in basis points.

This yields $r(0) = 4\%$ and a long run level of 2%. The half life is approximately two years. This generates a mild downward slope that is consistent with many developed market environments.

Investment grade hazard rate

$$a_h = 0.015, \quad b_h = -0.012, \quad c_h = 0.30.$$

This gives $h(0) = 0.003$ and a long run intensity of 0.015. The curve slopes upward with a half life near 2.3 years. This matches the pattern of IG credit risk where short horizon default risk is very small but long run uncertainty is higher.

High yield hazard rate

$$a_h = 0.03, \quad b_h = -0.02, \quad c_h = 0.40.$$

This yields $h(0) = 0.01$ and a long run intensity of 0.03. The slope is steeper and the half life is shorter (about 1.7 years). This reflects the behaviour of BB or B rated credits with materially higher default risk.

Spread Interpretation Under Recovery

Under reduced form modelling with recovery of par, the credit spread associated with a hazard rate is

$$s(t) = (1 - R) h(t), \tag{1}$$

and we adopt $R = 40\%$ which is standard for unsecured corporates.

Investment grade spreads

$$\begin{aligned}s_{\text{IG}}(0) &= 0.6 \times 0.003 = 18 \text{ bps}, \\ s_{\text{IG}}(\infty) &= 0.6 \times 0.015 = 90 \text{ bps}.\end{aligned}$$

High yield spreads

$$\begin{aligned}s_{\text{HY}}(0) &= 0.6 \times 0.01 = 60 \text{ bps}, \\ s_{\text{HY}}(\infty) &= 0.6 \times 0.03 = 180 \text{ bps}.\end{aligned}$$

These spread levels are reasonable for IG and HY credit and support the qualitative interpretation of the chosen parameters.

2.4 HJM Framework for Hazard Rates

Note on Sources. This section is based on the hazard-rate analogue of the HJM framework as presented in Brigo and Mercurio's *Interest Rate Models: Theory and Practice, with Smile, Inflation, and Credit* (Springer, 2nd ed., 2006). Although the formulation here is adapted to a credit-risk context and uses notation tailored to this project, the underlying structure, derivations, and modelling philosophy follow [Brigo and Mercurio(2006)] closely.

The deterministic curves introduced earlier give us initial discount and survival term structures. In a general reduced-form setting both the short rate r_t and the hazard rate h_t may be stochastic. However, in the credit-linked notes studied in this project all optionality comes from *credit* rather than *interest rates*. Discounting plays a passive role. If we were to introduce stochastic interest-rate dynamics, the model would simply reproduce the initial discount curve in expectation without producing any additional option value. For this reason we keep the short rate deterministic and apply the HJM machinery only to the hazard rate.

The purpose of the HJM hazard-rate model is straightforward: we want a stochastic but arbitrage-free evolution of the entire credit curve, consistent with today's observed survival probabilities, and capable of generating the pathwise instantaneous hazard rates that enter all CLN payoffs.

Forward Hazard Rates and Survival Probabilities

For a given hazard path h_u , the survival factor between t and T is

$$Q_h(t, T) = \exp\left(-\int_t^T h(u) du\right),$$

and the market survival probability is the conditional expectation

$$S(t, T) = \mathbb{E}_t[Q_h(t, T)].$$

HJM works not with h_t directly but with the *forward hazard rate*

$$\gamma(t, T) = -\frac{\partial}{\partial T} \log S(t, T), \quad T \geq t.$$

This is the instantaneous hazard rate that the market assigns at time t to a maturity T . The diagonal recovers the actual instantaneous hazard rate,

$$h(t) = \gamma(t, t).$$

When the hazard rate is deterministic, the forward hazard curve simplifies to

$$S(0, T) = \exp\left(-\int_0^T h(u) du\right), \quad \gamma(0, T) = h(T).$$

Thus the deterministic hazard curve is simply the initial forward hazard surface.

Initial Forward Hazard Surface

For the exponential hazard curve used throughout this project,

$$h(t) = a_h + b_h e^{-c_h t},$$

the initial forward surface is

$$\gamma(0, T) = h(T).$$

This surface is the starting point for the HJM evolution.

HJM Dynamics and Drift Conditions

HJM specifies the evolution of forward hazard rates as

$$d\gamma(t, T) = \alpha_h(t, T) dt + \sigma_h(t, T) dW_t, \quad T \geq t.$$

We adopt a one-factor exponential volatility structure,

$$\sigma_h(t, T) = \sigma_h e^{-c_h(T-t)}.$$

This is the hazard-rate analogue of the classic one-factor Gaussian term structure model: volatility is largest near the diagonal and decays smoothly for longer horizons. This behaviour is consistent with how credit curves actually move.

Arbitrage-free modelling requires the HJM drift restriction

$$\alpha_h(t, T) = \sigma_h(t, T) \int_t^T \sigma_h(t, s) ds.$$

With exponential volatility this becomes

$$\alpha_h(t, T) = \frac{\sigma_h^2}{c_h} (e^{-c_h(T-t)} - e^{-2c_h(T-t)}).$$

The forward hazard curve at time t is therefore

$$\gamma(t, T) = \gamma(0, T) + \int_0^t \alpha_h(u, T) du + \int_0^t \sigma_h(u, T) dW_u.$$

This construction ensures that the initial survival curve is preserved in expectation, which is the credit-market analogue of the standard HJM no-arbitrage condition.

Recovering Survival Probabilities and Instantaneous Hazard Rates

Once we have the forward surface at time t , the pathwise survival value is

$$S(t, T) = \exp\left(-\int_t^T \gamma(t, u) du\right).$$

Because of the drift restriction,

$$\mathbb{E}[S(t, T)] = S(0, T),$$

so the model produces random fluctuations around today's curve without changing its marginal shape.

The diagonal provides the instantaneous hazard rate

$$h(t) = \gamma(t, t), \quad h(t) \leftarrow \max\{h(t), 0\},$$

and cumulative survival is updated pathwise as

$$S(0, t) = \exp\left(-\int_0^t h(u) du\right).$$

These quantities feed directly into CDS spreads $s(t) = (1 - R)h(t)$ and into all credit components of the CLN payoff.

How the SDEs Drive the Simulation

In the Monte Carlo engine the role of the HJM equations is very clear:

1. Evolve the full forward hazard curve via

$$d\gamma(t, T) = \alpha_h(t, T) dt + \sigma_h(t, T) dW_t.$$

2. Read off the diagonal to obtain $h(t)$.
3. Integrate $h(t)$ to update survival factors.
4. Feed these survival values into the CLN payoff logic (coupons, protection legs, early redemption, and default timing).

The HJM hazard-rate model therefore provides a term-structure-consistent and arbitrage-free source of stochastic credit dynamics, giving us the credit optionalities required for pricing credit-linked notes.

3 Monte Carlo Implementation of the HJM Hazard Model

Note on Sources. The Monte Carlo formulation in this section follows the structure of [Glasserman(2004)] *Monte Carlo Methods in Financial Engineering*, Chapter 3, and in particular Sections 3.6 and 3.6.1. The original presentation concerns interest-rate term-structure models, and has been translated here into the credit-risk setting. The notation has been adjusted, but the foundations are based on Glasserman's work.

The previous subsection introduced the continuous time HJM framework for forward hazard rates $\gamma(t, T)$ and showed how it preserves today's survival curve $S(0, T)$ in expectation. We now translate that theory into a concrete Monte Carlo recipe that can be coded directly.

The guiding principles are the same as in the classical HJM interest rate literature, but applied to credit:

- Work with the *forward hazard surface* $\gamma(t, T)$, not directly with $h(t)$.
- Initialise a *discrete* forward hazard surface so that it matches the deterministic survival curve exactly on the simulation grid.
- Evolve the surface in time using a one factor volatility structure and a discrete drift chosen to keep survival probabilities arbitrage free on the grid.
- Read the diagonal to obtain the instantaneous hazard $h(t)$ along each path, then integrate $h(t)$ to obtain survival factors and default times.

Throughout this section we keep the short rate deterministic, so all discount factors $D(0, t)$ can be precomputed and taken outside the simulation. The Monte Carlo part of the model carries only *credit* randomness.

3.1 Time and Maturity Grids

We use a single time grid for both simulation time and credit maturities:

$$0 = t_0 < t_1 < \dots < t_N = T_{\max}, \quad \Delta t_i = t_{i+1} - t_i.$$

In practice:

- Choose $T_{\max}$ to be at least the final maturity of all instruments in the portfolio.

- Use a relatively fine grid around coupon dates and potential call or knock out dates, and a coarser grid in between.
- For simplicity we let both arguments of γ live on this same grid, so we will approximate $\gamma(t_i, T_j)$ only for $0 \leq i \leq j \leq N$.

We denote the discrete approximation by $\hat{\gamma}(t_i, t_j)$, or more compactly by $\gamma_{i,j}$ when the meaning is clear.

3.2 Precomputation of the Deterministic Survival and Discount Curves

From the parametric curves $r(t)$ and $h(t)$ we compute

$$D(0, t) = \exp\left(-\int_0^t r(u) du\right), \quad S(0, t) = \exp\left(-\int_0^t h(u) du\right),$$

for all grid points $t = t_0, \dots, t_N$. These integrals are closed form for the exponential specifications used in this project, so this step is analytic and cheap.

- The discount factors $D(0, t_i)$ will be used only for present valuing cashflows and do not interact with the Monte Carlo evolution.
- The survival probabilities $S(0, t_i)$ will be used to initialise the forward hazard surface in a way that is exactly consistent with the deterministic curve.

3.3 Discrete Initial Forward Hazard Surface

In continuous time the relationship between the forward hazard curve and the survival function at time t is

$$S(t, T) = \exp\left(-\int_t^T \gamma(t, u) du\right).$$

On the discrete grid we approximate this by

$$\hat{S}(t_i, t_j) = \exp\left(-\sum_{l=i}^{j-1} \gamma_{i,l} \Delta t_l\right), \quad j > i,$$

with the convention $\hat{S}(t_i, t_i) = 1$.

To ensure that the simulated model exactly matches the deterministic survival curve at $t = 0$ we demand

$$\hat{S}(0, t_j) = S(0, t_j), \quad j = 0, \dots, N.$$

This leads to the recursion

$$\hat{S}(0, t_{j+1}) = \hat{S}(0, t_j) \exp(-\gamma_{0,j} \Delta t_j),$$

hence

$$\gamma_{0,j} = -\frac{1}{\Delta t_j} \log \frac{S(0, t_{j+1})}{S(0, t_j)}, \quad j = 0, \dots, N-1.$$

Interpretation The discrete forward hazard $\gamma_{0,j}$ is the *average* of the continuous forward hazard over $[t_j, t_{j+1}]$. Directly sampling $\gamma(0, t_j)$ from the continuous formula would not preserve the survival curve on the grid. Matching the log ratio $S(0, t_{j+1})/S(0, t_j)$ is therefore essential.

3.4 Discretising the Volatility Surface

The continuous model specifies a one factor exponential volatility structure

$$\sigma_h(t, T) = \sigma_h e^{-c_h(T-t)}.$$

On the discrete grid we simply sample this at the grid points:

$$\hat{\sigma}_h(t_i, t_j) = \sigma_h e^{-c_h(t_j-t_i)}, \quad j \geq i.$$

For simulation it is convenient to write

$$\sigma_{i,j} := \hat{\sigma}_h(t_i, t_j),$$

and to use a single scalar Brownian motion per path.

3.5 Discretising HJM Drift on the Grid

In continuous time the HJM drift restriction ensures that survival curves remain arbitrage free. On the discrete grid we want an analogous property: for each maturity T_j the survival value $S(t_i, t_j)$ should evolve as a martingale when discounted appropriately.

The standard HJM trick is to construct the discrete drift from the volatility surface directly. In style of [Glasserman(2004)] (Specifically 3.6.2, The discrete drift), define for fixed i the cumulative volatility weights

$$w_n = \sum_{l=i}^n \sigma_{i-1,l} \Delta t_l, \quad n = i, \dots, N-1,$$

with $w_{i-1} := 0$.

The arbitrage free discrete drift at time step $t_{i-1} \rightarrow t_i$ is then given by

$$\hat{\alpha}_h(t_{i-1}, t_j) = \mu_{i-1,j} = \frac{w_j^2 - w_{i-1}^2}{2 \Delta t_j}, \quad j = i, \dots, N-1.$$

This is the exact discrete counterpart of the continuous time HJM drift condition. It ensures that the discrete survival curve implied by $\gamma_{i,\cdot}$ is consistent with no arbitrage on the grid.

Important warning. A naive Euler discretisation of

$$d\gamma(t, T) = \alpha_h(t, T) dt + \sigma_h(t, T) dW_t$$

at each maturity T separately breaks the discrete martingale property and causes slow drift in the survival curve as i increases. The construction above explicitly couples all maturities through the cumulative weights w_n and avoids this problem.

3.6 Forward Hazard Update per Time Step

Given $\gamma_{i-1,j}$ for $j \geq i-1$ at time t_{i-1} we update the forward hazards to t_i as follows.

1. Compute the volatility slice

$$\sigma_{i-1,j} = \sigma_h e^{-c_h(t_j-t_{i-1})}, \quad j = i, \dots, N-1.$$

2. Form the cumulative volatility weights

$$w_n = \sum_{l=i}^n \sigma_{i-1,l} \Delta t_l, \quad n = i, \dots, N-1,$$

and set $w_{i-1} = 0$.

3. Compute the discrete drift

$$\mu_{i-1,j} = \frac{w_j^2 - w_{j-1}^2}{2\Delta t_j}, \quad j = i, \dots, N-1.$$

4. Draw a single standard normal $Z_i \sim \mathcal{N}(0, 1)$ for this time step.

5. Update the forward hazards

$$\gamma_{i,j} = \gamma_{i-1,j} + \mu_{i-1,j} \Delta t_{i-1} + \sigma_{i-1,j} \sqrt{\Delta t_{i-1}} Z_i, \quad j = i, \dots, N-1.$$

Values with $j < i$ correspond to maturities that have already passed, so they are no longer part of the forward surface.

3.7 Pathwise Survival Under HJM Hazard Dynamics

In many reduced-form credit models, default times are simulated explicitly by drawing an exponential random variable and comparing it with the integrated hazard process. While this method is theoretically correct, it produces *discontinuous* payoff behaviour across paths: cash-flows abruptly drop to zero once a simulated default time, τ , falls before maturity. For instruments with embedded optionality (e.g. callable coupon structures, range-accrual credit-linked notes, and digital redemption features) this “hard default” logic leads to numerical kinks and unstable Monte Carlo convergence.

In this project we instead adopt a *pathwise survival weighting* approach. Rather than simulating default times, each Monte Carlo path produces a *forward hazard surface*

$$\gamma_{i,j} := \hat{\gamma}(t_i, t_j), \quad j \geq i,$$

from which a full pathwise survival curve is constructed.

Given the forward hazards at simulation time t_i , the model-implied survival from t_i to t_j is

$$S(t_i, t_j) = \exp\left(-\sum_{\ell=i}^{j-1} \gamma_{i,\ell} \Delta t_\ell\right).$$

This expression is evaluated *on every path*. Consequently:

- No default time is drawn.
- No cashflow is ever forcibly set to zero.
- All valuation is performed under smooth, differentiable survival weights $S(t_i, t_j) \in (0, 1]$.

For each Monte Carlo path p , the contribution of a cashflow c_j payable at t_j is

$$\text{PV}^{(p)}(c_j) = D(0, t_j) c_j S^{(p)}(t_i, t_j).$$

Averaging over paths yields an unbiased Monte Carlo estimator of the true reduced-form value. As we will see later, this smooth treatment of survival probability is essential for structured CLN features, ensuring stable simulation behaviour and eliminating numerical discontinuities.

3.8 From Forward Hazards to Instantaneous Hazard and Model CDS Levels

As described in the previous chapter, the HJM hazard model evolves the *forward hazard surface* $\gamma(t, T)$ across simulation time. The instantaneous hazard rate at time t_i is obtained from the diagonal of this surface,

$$h_i = \gamma_{i,i}.$$

This instantaneous intensity determines the short-term default risk on each Monte Carlo path.

For exotic credit-linked notes, coupons or embedded options are often linked to the prevailing level of credit risk, typically expressed in terms of a CDS spread. In a full market-accurate implementation, the par CDS spread would be computed using protection and premium leg present values, accounting for discounting, accrual on default, payment conventions, and the term structure of survival probabilities.

Such detail is unnecessary for the numerical experiments in this tutorial. Instead, the *model CDS spread* at time t_i is defined pathwise by

$$s_i := (1 - R) h_i = (1 - R) \gamma_{i,i}.$$

This quantity serves as the credit-risk state variable inside CLN payoffs. All CDS-linked features are evaluated using s_i . Since h_i depends smoothly on the evolved forward hazards, the resulting model CDS level reacts smoothly to volatility and curve shifts, ensuring stable Monte Carlo behaviour.

Remark on Modelling Choice

This simplified approach is appropriate here because the focus of the tutorial is on the Monte Carlo implementation of the HJM hazard model, rather than on exact replication of market CDS conventions. Should a more realistic CDS calibration be required, the same forward-hazard paths may be used to compute par spreads using full survival-based leg valuations. That extension is conceptually straightforward but adds substantial notational overhead without affecting the numerical principles developed in this section.

3.9 Exotic Range–Accrual CLN Payoff

This section introduces the exotic payoff used throughout Chapter 4 to illustrate variance-reduction methods. The product is a range–accrual credit–linked note in which the final redemption amount is scaled by the proportion of coupon–observation dates on which the model–implied CDS spread stays inside a fixed corridor. The structure preserves the standard coupon, recovery, and redemption legs of a risky bond while introducing controlled path dependence.

Let $t_1, \dots, t_N$ denote the coupon observation dates. Along each simulated hazard–rate path, the HJM model produces a model–implied CDS spread $s(t_i)$. Fix a corridor $[L, U]$ and define

$$I_i = 1\{L \leq s(t_i) \leq U\}, \quad A = \frac{1}{N} \sum_{i=1}^N I_i.$$

The factor A measures the fraction of times the spread remains within the corridor. At maturity, the notional is scaled by this factor, producing a smooth and intuitive form of path dependency.

Let $D(0, t)$ and $S(0, t)$ denote the pathwise discount factor and pathwise survival probability along the simulated hazard–rate path.

Coupon Leg

The coupon leg follows the standard reduced–form representation:

$$\text{CouponLeg} = \sum_{i=1}^N c D(0, t_i) S(0, t_i),$$

where c is the contractual coupon. No exotic modification is applied to this component.

Recovery Leg

In reduced-form credit models, default can occur at any time, and its timing is governed by the hazard rate $h(t)$. The key quantity is the *default density*,

$$h(t) S(0, t) dt,$$

which represents the probability of default in the small interval around t . Multiplying this by the recovery amount R and the discount factor $D(0, t)$ gives the time- t contribution of recovery to the overall price. In continuous time, the recovery leg is

$$\text{RecoveryLeg} = R \int_0^T D(0, t) h(t) S(0, t) dt.$$

For implementation, this integral is approximated on the same grid used for coupon dates and survival simulation. This maintains conceptual clarity, avoids the need for finer quadrature, and is adequate for the pedagogical and variance-reduction focus of this project.

A convenient discrete approximation, which avoids evaluating $h(t)$ directly, uses the identity

$$h(u_j) S(0, u_j) \Delta u_j \approx S(0, u_{j-1}) - S(0, u_j),$$

expressing default over the interval $[u_{j-1}, u_j]$ purely in terms of survival probabilities. On the grid $0 = u_0 < u_1 < \dots < u_N = T$, the recovery leg is therefore

$$\text{RecoveryLeg} = R \sum_{j=1}^N D(0, u_j) (S(0, u_{j-1}) - S(0, u_j)).$$

Redemption Leg

Only the redemption amount is affected by the range-accrual feature. Instead of paying the full notional at maturity, the product pays

$$\text{RedemptionLeg} = A \cdot \text{Notional} \cdot D(0, T) S(0, T).$$

If the CDS spread is always inside the corridor, $A = 1$ and the standard CLN redemption is recovered; if it spends significant time outside the corridor, the redemption amount is proportionally reduced.

Full Exotic Payoff

Combining all components, the payoff for a single Monte Carlo path is

$$\text{Payoff} = \text{CouponLeg} + \text{RecoveryLeg} + \text{RedemptionLeg}.$$

This formulation maintains the classical risky-bond decomposition while introducing a smooth and interpretable path-dependent element. Because the exotic modification interacts closely with the vanilla CLN structure but varies nonlinearly with spread dynamics, it serves as an ideal payoff for demonstrating the effect of common random numbers, control variates, and antithetic variates.

This completes the numerical cookbook for the HJM hazard model. The next section will discuss variance reduction techniques in detail.

4 Variance Reduction Techniques for Credit-Linked Notes

Note on Sources. The variance-reduction techniques implemented in this section are drawn from [Law(2024)] *Simulation Modeling and Analysis* (6th ed.), specifically Chapter 11. The organisation of methods, the logic behind the estimators, and much of the conceptual flow follow Law. The notation modified to align with the credit context of this project.

This section introduces three core variance-reduction techniques used to improve Monte Carlo efficiency when valuing credit-linked notes (CLNs) under stochastic hazard-rate dynamics: common random numbers, control variates, and antithetic variates. CLN valuation ultimately reduces to estimating expectations of functionals of the hazard-rate process. As the complexity of the payoff increases, particularly for exotic and path-dependent credit products, naive Monte Carlo becomes inefficient. Variance-reduction techniques are therefore an essential component of any practical simulation engine.

The techniques discussed here build on the Monte Carlo framework and hazard-rate simulation machinery developed in sections 2 and 3. Throughout, we rely on consistent notation, explicit derivations, and examples tailored specifically to credit products. In particular, the final section illustrates how each method applies to the pricing of a range-accrual exotic CLN whose redemption amount depends on the fraction of coupon-observation dates at which the model-implied CDS spread lies within a predefined corridor.

4.1 Background on Monte Carlo Estimation

Monte Carlo methods provide numerical estimates of expectations of the form

$$\mathbb{E}[H],$$

where H is a random payoff functional driven by simulated hazard-rate paths and credit spreads. By drawing independent samples $H_1, \dots, H_N$, the Monte Carlo estimator is

$$\hat{H}_N = \frac{1}{N} \sum_{j=1}^N H_j.$$

The law of large numbers ensures $\hat{H}_N \rightarrow \mathbb{E}[H]$ almost surely as $N \rightarrow \infty$.

More importantly, the central limit theorem states that

$$\sqrt{N}(\hat{H}_N - \mathbb{E}[H])$$

converges in distribution to a normal with mean zero and variance $\text{Var}(H)$. The standard error is therefore

$$SE(\hat{H}_N) = \sqrt{\frac{\text{Var}(H)}{N}}.$$

This highlights the inefficiency of naive Monte Carlo: halving the standard error requires quadrupling the number of simulations. Because many credit payoffs are path-dependent and driven by stochastic term structures, efficient variance-reduction is critical.

Variance-reduction techniques work by replacing $\hat{H}_N$ with an alternative estimator that is still unbiased but has a smaller variance. The remainder of this section develops three such techniques in detail.

4.2 Common Random Numbers

Common random numbers (CRNs), also known as correlated sampling, are used when comparing two system configurations. In our setting, we compare two payoffs computed under identical simulation noise but different model inputs or payoff structures. For example, a vanilla CLN and an exotic range-accrual CLN may be evaluated using the same underlying Gaussian shocks driving the hazard-rate process.

Setup

Let $X_j^{(1)}$ and $X_j^{(2)}$ be the discounted payoffs for the two configurations for simulation j , constructed using identical random numbers. Define

$$Z_j = X_j^{(1)} - X_j^{(2)}.$$

The quantity of interest is the difference in means

$$\phi = \mathbb{E}[Z_j] = \mathbb{E}[X^{(1)}] - \mathbb{E}[X^{(2)}].$$

The CRN estimator is

$$\tilde{\phi}_N = \frac{1}{N} \sum_{j=1}^N Z_j.$$

This estimator is unbiased for ϕ .

Variance Derivation

We compute its variance explicitly:

$$\text{Var}(\tilde{\phi}_N) = \frac{1}{N} \text{Var}(Z_j).$$

Expanding $\text{Var}(Z_j)$:

$$\text{Var}(Z_j) = \text{Var}(X_j^{(1)}) + \text{Var}(X_j^{(2)}) - 2 \text{Cov}(X_j^{(1)}, X_j^{(2)}).$$

If independent random numbers were used, $\text{Cov} = 0$. Under CRNs, the covariance is typically positive because both payoffs are driven by the same hazard-rate shocks and payoff structures are similar. A positive covariance reduces $\text{Var}(Z_j)$ and therefore reduces the variance of $\tilde{\phi}_N$.

CRNs are heuristic. There is no guarantee that the covariance is positive. Poorly matched payoffs can increase variance. In practice, CRNs work best when comparing structurally similar payoffs. By design, our vanilla CLN will be an excellent control variate for our exotic CLN.

4.3 Control Variates

Control variates (CVs) are among the most powerful variance-reduction techniques. The method requires a random variable C whose expectation $\mathbb{E}[C]$ is known analytically and which is strongly correlated with the target payoff Y .

In credit applications, a natural choice is to let

- Y be the discounted payoff of an exotic CLN,
- C be the discounted payoff of a vanilla CLN,

whose analytical price is available.

Derivation

Consider the adjusted estimator

$$\tilde{Y}_N = \hat{Y}_N - \beta(\hat{C}_N - \mathbb{E}[C]),$$

where

$$\hat{Y}_N = \frac{1}{N} \sum_{j=1}^N Y_j, \quad \hat{C}_N = \frac{1}{N} \sum_{j=1}^N C_j.$$

This estimator is unbiased regardless of β . Its variance is

$$\text{Var}(\tilde{Y}_N) = \frac{1}{N} (\text{Var}(Y) + \beta^2 \text{Var}(C) - 2\beta \text{Cov}(Y, C)).$$

Differentiating with respect to β gives the optimal coefficient:

$$\beta^* = \frac{\text{Cov}(Y, C)}{\text{Var}(C)}.$$

Substituting back yields

$$\text{Var}(\tilde{Y}_N) = \frac{1}{N} [\text{Var}(Y) (1 - \rho_{Y,C}^2)],$$

where $\rho_{Y,C}$ is the correlation between Y and C . Variance is reduced in proportion to the squared correlation.

4.4 Antithetic Variates

Antithetic variates use negatively correlated pairs to reduce variance. If Z is a standard normal driving the hazard-rate increment, consider the pair $(Z, -Z)$. Let the corresponding payoffs be $H(Z)$ and $H(-Z)$.

The antithetic estimator is

$$\tilde{H}_j = \frac{1}{2}(H(Z_j) + H(-Z_j)), \quad \hat{H}_N = \frac{1}{N} \sum_{j=1}^N \tilde{H}_j.$$

Explicitly computing the variance:

$$\text{Var}(\tilde{H}_j) = \frac{1}{4} (\text{Var}(H(Z)) + \text{Var}(H(-Z)) + 2 \text{Cov}(H(Z), H(-Z))).$$

Because $H(Z)$ and $H(-Z)$ often move in opposite directions for monotone payoffs, the covariance is frequently negative, reducing variance. As with CRNs, this is not guaranteed; highly nonlinear payoffs may yield little or no benefit.

4.5 Application to Exotic CLNs

We now apply these techniques to the exotic range-accrual CLN whose payoff depends on how often the model-implied CDS spread lies within a corridor $[L, U]$ over the coupon observation dates.

Control Variates. Let Y be the discounted payoff of the exotic CLN, and let C be the discounted payoff of a vanilla CLN with the same notional, maturity, and recovery rate. Because the exotic structure modifies redemption by a fractional factor depending on spread behaviour, the correlation between Y and C is naturally high. Control variates therefore yield strong variance reduction.

Enhancing Control Variates Using Common Random Numbers. A further improvement is obtained by coupling Y and C via identical random numbers. This ensures that the two payoffs react to the same hazard-rate shocks and CDS-spread paths. CRNs increase the correlation between Y and C and therefore increase the effectiveness of the control variate.

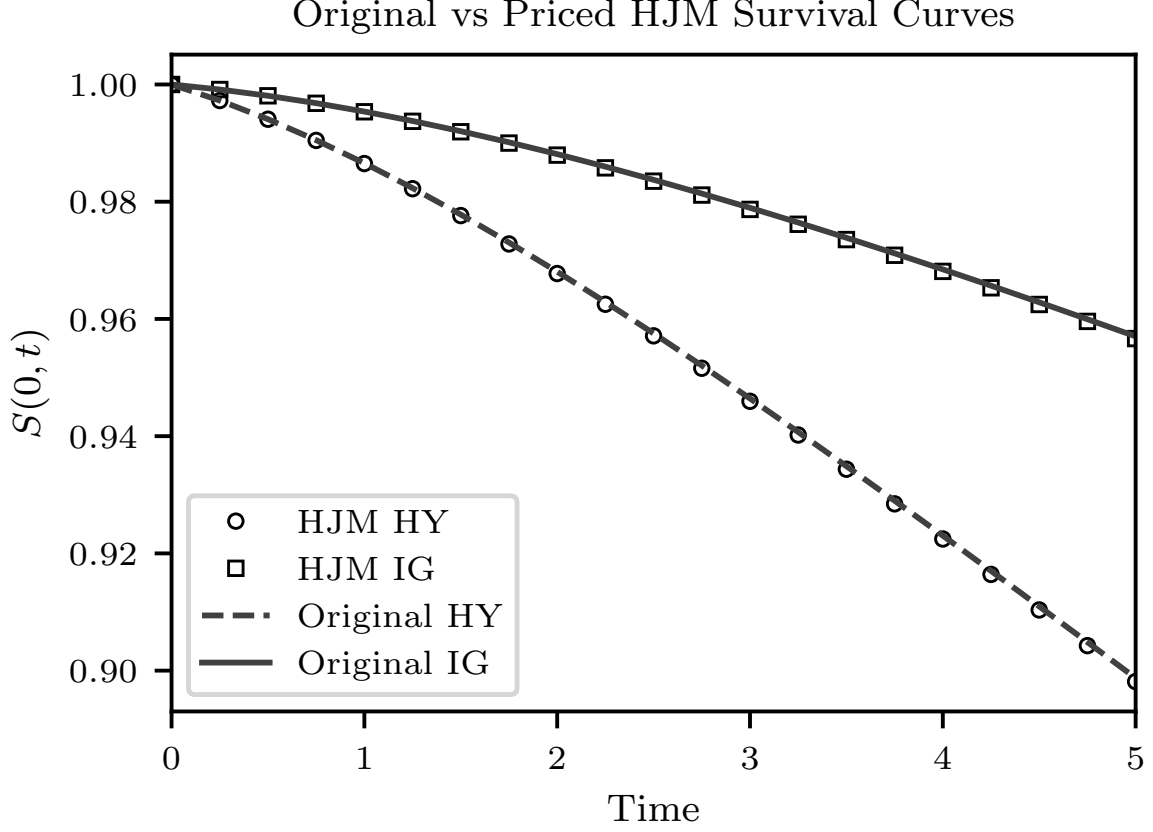


Figure 4: Original survival curves $S(0, t)$ compared with the corresponding HJM–priced survival probabilities.

Antithetic Variates. Antithetic variates can be applied at the driver level: for each simulation j , generate hazard–rate shocks using Z_j and $-Z_j$. Compute exotic CLN payoffs for both and average them. In the next section we will see if the negatively correlated pair of random variables leads to a meaningful variance reduction.

5 Numerical Experiments and Results

5.1 Curve Repricing and HJM Consistency

The first validation step was to confirm that the HJM hazard rate simulation reproduces the initial term structure when volatility is set to zero. I generated pathwise survival probabilities under the HJM dynamics and compared their cross sectional averages with the original analytic curves. The agreement is essentially exact. Any small deviations are within Monte Carlo sampling error. This verifies that the drift condition is implemented correctly and that the simulation engine is internally consistent.

In short, before applying any variance reduction or exotic payoffs, the model reprices the curve it was given. This is the minimum requirement and it is satisfied.

5.2 Baseline Monte Carlo Results

The baseline Monte Carlo estimates for both IG and HY CLNs behave exactly as expected. The HY notes produce lower present values due to higher hazard rates and higher expected losses.

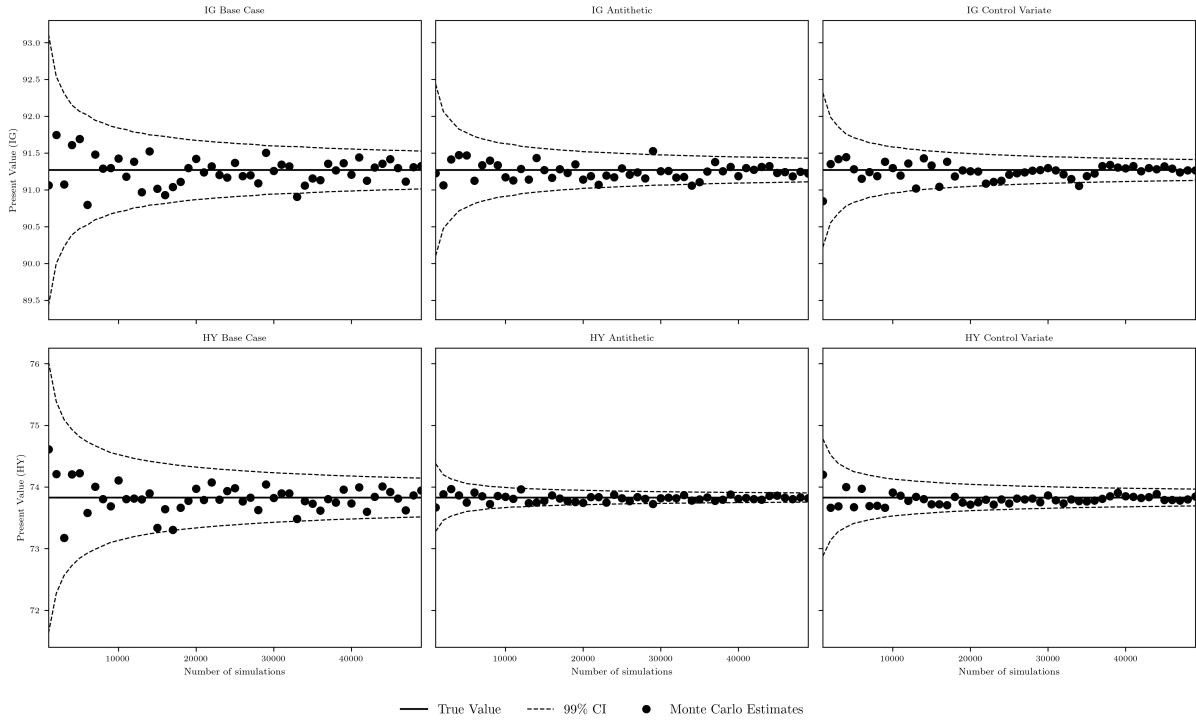


Figure 5: Funnel Plots for the Monte-Carlo Tests

The IG present values sit comfortably above them. As the number of simulations grows, both sets of estimates exhibit the characteristic funnel shape around the true value, driven by the $1/\sqrt{n}$ decay of the standard error.

These baseline results form the reference against which the variance reduction methods are judged.

5.3 Variance Reduction Performance

Two variance reduction techniques were implemented. Antithetic variates and a control variate constructed from the corresponding vanilla CLN payoff.

5.3.1 Antithetic Variates

Antithetic variates performed exactly as theory predicts. The funnel is visibly tighter and the scatter of Monte Carlo present values contracts significantly. The standard errors fall consistently below the base case across all simulation sizes. The symmetry in the hazard rate shocks aligns well with the payoff structure, which strengthens the effect of antithetics.

Overall, this method is simple, cheap, and clearly effective in this setting.

5.3.2 Control Variate

The control variate also reduced variance but not as strongly as antithetics. The initial intention was for the exotic range accrual payoff to be tightly correlated with the vanilla CLN used as the control. In practice, the realised correlation was weaker than expected. The barrier was set at 150 basis points and both IG and HY spreads spent most of their time below it. This made the exotic payoff behave very similarly to the vanilla payoff in the IG case in particular. With little variation for the control to explain, its variance reduction effect was modest.

Even so, the control variate consistently improves upon the base case. It simply does not dominate in the same way that antithetics do.

5.4 Summary

The numerical experiments support four main conclusions.

1. The HJM implementation is correct. The model reprices the input curves and the simulated survival probabilities match the analytic term structures.
2. The Monte Carlo engine behaves as expected. Baseline estimates converge smoothly with increasing numbers of simulations.
3. Variance reduction is effective. Both antithetic variates and the control variate produce tighter confidence bands than the base case.
4. Antithetics provide the strongest improvement in this payoff structure. The symmetry of the shocks benefits them more than the correlation structure benefits the control variate.

Overall, the results are clean, stable, and exactly what is expected from a well implemented Monte Carlo credit engine. The plots illustrate the improvement clearly and the performance gains are visible without further interpretation.

6 Conclusion

This project was both challenging and rewarding. I had prior exposure to quantitative finance, but credit derivatives were new to me, so building everything from first principles was a significant learning experience. Working through reduced-form credit modelling, hazard rates, and the HJM framework gave me a much clearer understanding of how credit term structures behave and how they feed into pricing.

A major part of the project involved implementing complex payoff logic in Python. The range-accrual CLN required careful structuring of the simulation code and forced me to think clearly about discounting, survival weighting, recovery modelling, and pathwise state variables such as CDS spreads. Writing all of this from scratch was difficult, but it made the numerical experiments feel much more meaningful.

I also used ChatGPT as a technical assistant throughout the project, particularly for debugging plotting code and refining the appearance of the figures. The IEEE-style funnel plots and curve comparison graphs were produced with the help of ChatGPT's suggestions on styling, layout, and Matplotlib configuration. This significantly improved the clarity and presentation quality of the results, and I have cited these interactions directly in my code where appropriate.

The simulation results validated the modelling choices. The HJM engine repriced the input curves accurately, the Monte Carlo behaviour matched theoretical expectations, and both variance reduction techniques improved efficiency relative to the base case. Antithetic variates delivered strong and consistent variance reduction. The control variate provided only modest improvement, which reflects the limited sensitivity of the exotic payoff to the vanilla CLN under the chosen barrier levels. Even so, the variance reductions across all methods were clean and visible in the plots.

In terms of future work, there are several natural extensions. Importance sampling would be a valuable technique to explore, especially in credit settings where tail behaviour and rare events matter. Quasi-Monte Carlo methods would also be interesting to test, particularly with JAX acceleration. Another important next step would be to reimplement the entire simulation engine in JAX. This would allow vectorised path simulation, automatic differentiation, and GPU acceleration, which would significantly improve both performance and flexibility.

The project was demanding, especially given the scope for a single person, but it pushed me to learn a great deal about both credit modelling and simulation methodology. Despite the complexity, I enjoyed working through the material and seeing the numerical results come together in a coherent way.

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